Deep Learning for Medical Imaging: Out-of-Distribution Histopathology Image Classification

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Abstract

We present a solution to the out-of-distribution histopathology image classification challenge, where models trained on patches from three clinical centers must generalize to a fully held-out fifth center. Starting from a lightweight DINOv2 embedding baseline with per-center normalization, we identify the failure modes of frozen-feature approaches and progressively refine our solution toward end-to-end fine-tuning of KimiaNet equipped with Macenko stain augmentation and Low-Rank Adaptation (LoRA). Macenko augmentation directly addresses the stain-concentration domain gap by simulating realistic inter-center variability at training time, while LoRA constrains fine-tuning to a low-rank subspace to prevent overfitting to source-center styles. The final model achieves 0.95 accuracy on the held-out validation center and 0.95 on the Kaggle test set, compared to 0.90 for ImageNet-pretrained baselines.

Keywords: domain generalization, histopathology, stain augmentation, LoRA, KimiaNet

1. Introduction

Computational pathology relies on models trained at one clinical site being deployed at another, yet histopathology slides stained and scanned across hospitals exhibit strong visual variability. This *domain shift* stems from differences in staining protocols, reagent lots, and scanner hardware, and severely degrades classifier performance on unseen centers.

Models are trained on patches from three centers (C1, C2, C3), validated on C4, and evaluated on a held-out fifth center (C5) — a strict *leave-one-domain-out* generalization problem. Our solution targets the key failure modes of naive fine-tuning with three components: (i) **KimiaNet** (Riasatian et al., 2021), a histopathology-pretrained backbone; (ii) **Macenko stain augmentation** (Macenko et al., 2009), to simulate inter-center staining variability at training time; and (iii) **LoRA** (Hu et al., 2022), to regularize fine-tuning and limit overfitting to source-center styles.

2. Exploratory data analysis

To characterize the domain shift, we applied UMAP (cosine metric, 15 neighbours) to a 10 000-sample subset of frozen DINOv2 ViT-S/14 embeddings from the three training centers. Figure 1 shows that the centers occupy largely disjoint regions of the 2-D embedding, while the tumour/normal label boundary does not align with center membership. This confirms that center identity is a dominant source of variance in the feature space, directly motivating stain augmentation and a histopathology-specific backbone.

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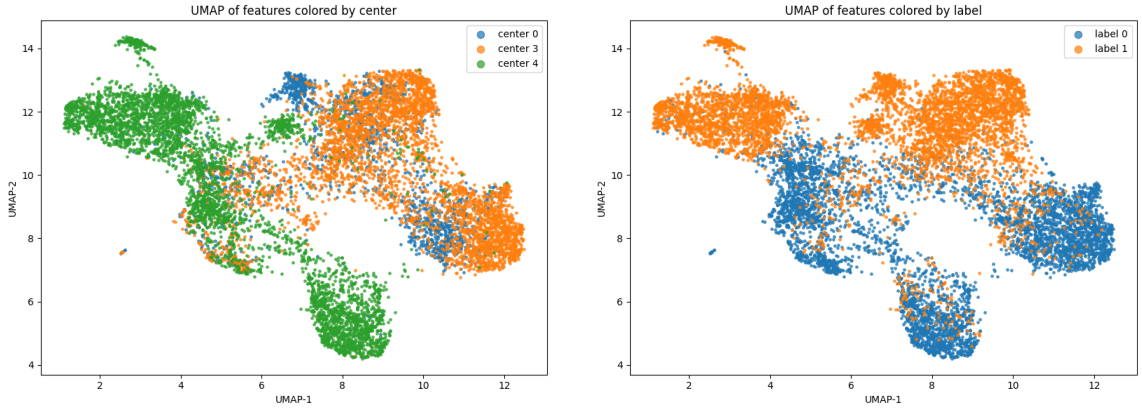

Figure 1: UMAP of DINOv2 embeddings coloured by center (left) and by label (right). Centers cluster separately; the label boundary does not follow center membership.

3. Architecture and Methodological Components

3.1. Backbone: KimiaNet

Motivation. KimiaNet (Riasatian et al., 2021) is a DenseNet-121 backbone pre-trained on over 240 000 TCGA diagnostic slides, giving representations attuned to tissue microstructure and nuclear morphology rather than scanner-specific artefacts.

Architecture. The backbone's final global average pooling layer produces a 1024-dimensional feature vector, which is fed into a linear classification head projecting to K classes:

$$f_\theta(\mathbf{x}) = W_{\text{cls}}\,\text{GAP}(\phi_{\text{KimiaNet}}(\mathbf{x})) + b_{\text{cls}}.$$

The full pipeline is illustrated in Figure 2.

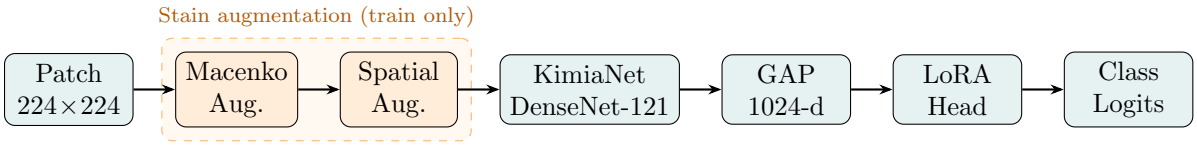

Figure 2: Overview of the final pipeline. Orange blocks are applied only at training time.

3.2. Stain Augmentation: Macenko Method

Motivation. Color jitter and standard photometric augmentations operate in RGB space and do not respect the physics of bright-field staining. Inter-center variability is primarily a *stain-concentration* effect: the same tissue stained with H&E at two hospitals looks similar structurally but differs systematically in hue and saturation. An augmentation that directly perturbs stain concentrations will therefore generate samples that faithfully simulate the domain gap, forcing the model to learn stain-invariant features.

Implementation. Following Howard et al. (2021), transformations are applied in order: (1) **Macenko augmentation**: multiplicative perturbation of the stain matrix, probability p_{MA}; (2) **spatial augmentations**: random flips and 90° rotations (all orientations are biologically valid); (3) **color jitter**; (4) **Gaussian blur** ($p = 0.2$, kernel 3); (5) **ImageNet normalization**.

The Macenko augmentor (Macenko et al., 2009) fits a 2×3 stain matrix $\mathbf{S}$ on each image via SVD of its optical density point cloud. The two principal stain vectors are independently perturbed by multiplicative Gaussian noise:

$$\tilde{\mathbf{S}}_i = \mathbf{S}_i \cdot \exp(\sigma_i \, \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(0, 1), \quad i \in \{1, 2\},$$

and the image is reconstructed as $\hat{\mathbf{x}} = I_o \exp(-\mathbf{C}\tilde{\mathbf{S}})$ where $\mathbf{C} = \mathbf{OD}\,\mathbf{S}^{\dagger}$ are the stain concentrations.

Discarded alternatives. Macenko stain *normalization* (mapping all patches to a consensus appearance) was tried but degraded out-of-distribution generalization compared to augmentation. HED Jitter (additive noise in HED stain-concentration space) was also tested as an alternative to Macenko but was not better than Macenko.

3.3. Fine-Tuning: Two-Stage LoRA

Motivation. Naive full fine-tuning of KimiaNet on three source centers risks catastrophic forgetting of generalizable histology features and overfitting to center-specific appearance. LoRA (Hu et al., 2022) constrains weight updates to a low-rank subspace:

$$W = W_0 + \Delta W = W_0 + BA, \quad B \in \mathbb{R}^{d \times r}, \; A \in \mathbb{R}^{r \times k}, \; r \ll \min(d, k),$$

acting as a structural regularizer that preserves pre-trained features while allowing task-specific adaptation.

Two-stage recipe. **Stage 1** freezes the entire backbone and trains only the classification head for a few epochs. This stabilizes the output distribution before any backbone modification. **Stage 2** injects LoRA adapters into the linear layers of the DenseNet and unfreezes the head, training both jointly. The frozen weights W_0 are never updated; only A, B, and the head are optimized.

4. Model Tuning and Comparison

4.1. Validation Procedure

During the initial embedding exploration (Section 4.2), we used **Leave-One-Center-Out CV (LOOCV)** over the three training centers, which is cheap when only a linear probe is retrained. Once we moved to end-to-end fine-tuning, LOOCV became impractical, so we switched to **C4 as a fixed held-out validation center**. C4 was never used for gradient updates; the final model is selected solely on C4 accuracy.

4.2. Development Journey

Act 1: DINOv2 embedding baseline. We began with frozen DINOv2 (Oquab et al., 2024) (ViT-S/14) embeddings and a linear probe evaluated via LOOCV, using per-center feature normalization to mitigate domain shift. This has a fundamental flaw: center identity is unknown at test time, and frozen natural-image features encode scanner-specific textures that normalization cannot remove. The approach failed to generalize and was discarded.

Act 2: End-to-end fine-tuning with stain augmentation. We switched to KimiaNet with Macenko stain augmentation, directly simulating inter-center variability during training. This yielded a substantial C4 improvement (Table 1); validation moved to the single held-out C4 center.

Act 3: LoRA regularization. Full fine-tuning showed a train–validation gap indicative of overfitting to source-center styles. LoRA adapters closed this gap and reduced trainable parameters to less than one million. TTA via stain perturbation was tried but did not improve C4 accuracy.

4.3. Ablation Study

Table 1 isolates the contribution of each component on C4. All variants share the same KimiaNet backbone and training budget.

Table 1: Ablation on validation center C4. Each row adds one component over the previous.

Configuration	**Train Acc.**	**C4 Acc.**
Head-only + stain augmentation	0.85	0.75
Full fine-tune, no augmentation	0.98	0.89
Full fine-tune + stain augmentation	0.97	0.93
LoRA fine-tune + stain augmentation	0.96	0.95

4.4. Preprocessing and Training Details

Input normalization. Images are normalized with ImageNet mean and standard deviation before being fed to the backbone, consistent with KimiaNet's pre-training protocol.

Class balancing. The training set is approximately balanced between the two classes; no re-weighting was applied.

4.5. Hyperparameters

Table 2: Hyperparameters for each training stage.

Hyperparameter	**Stage 1 (head)**	**Stage 2 (LoRA)**
Optimizer	AdamW	AdamW
Learning rate	1×10^{-3}	1×10^{-3}
Weight decay	1×10^{-4}	1×10^{-4}
Batch size	512	512
Epochs	1	10
LR schedule	Cosine	Cosine
LoRA rank r	—	8
LoRA α	—	16

4.6. Results and Comparison

Table 3: Final performance of our model versus baselines.

Model	**Train Acc.**	**C4 Acc.**	**Kaggle**
ResNet-50 (ImageNet), full FT	> 0.98	0.92	0.90
ViT-S/16 (DINO), full FT	> 0.98	0.91	0.90
KimiaNet, head-only + aug.	0.85	0.75	—
KimiaNet + LoRA + stain aug.	0.97	0.95	0.95

Discussion. Stain augmentation was the single most impactful component: omitting it limited C4 accuracy to 0.89 despite 0.98 training accuracy, illustrating the severity of stain-based distribution shift. LoRA added +0.02 C4 accuracy while reducing trainable parameters to $\sim$20% of full fine-tuning. ResNet-50 and ViT-S/16 baselines were competitive (0.91–0.92) but outperformed by KimiaNet, confirming the benefit of a histopathology-specific initialization.

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